



ML.NET:

The New Open Source

Machine Learning

Framework

from Microsoft

Machine learning is the new rage. It is indeed an exciting time when new frameworks and new tools are being developed constantly. This is a write up on Microsoft's foray into open source with a new offering—ML.NET.

The number of developers working on machine learning (ML) is growing fast. One of the main reasons for this is the availability of an increasing number of easy-to-use, powerful ML frameworks from many industry leaders such as Google, Microsoft, etc. ML.NET is a recent ML framework from Microsoft with cross-platform open source features. This article will introduce you to the features of the ML.NET framework, along with code examples.

Machine learning is the new superstar in cyberspace. It has applications across all domains and has solved problems that were earlier considered unsolvable. Prominent examples are image recognition, behaviour understanding, speaker neutral voice recognition with almost 100 per cent accuracy, etc.

Research in ML can be classified into two types. One is the actual algorithm development, and the other is exploring and building various applications using ML. The first one requires a deep understanding of various mathematical

concepts. The second one, however, doesn't require you to understand the inner working mechanisms of the algorithms. The applications part of ML is gaining popularity across a wide spectrum of developers. The primary catalyst for this is the availability of various ML and deep learning (DL) libraries such as TensorFlow, Scikit Learn, Caffe, etc.

ML.NET was released by Microsoft on May 7, 2018. It is written in C# and C++, and its important attributes are listed below.

- ML.NET is cross-platform; it is supported on Linux, Windows and MacOS.
- It is open source (<https://github.com/dotnet/machinelearning>).
- It can be made to co-work with TensorFlow, CNTK, etc.
- If you are a .NET developer, you will find ML.NET very familiar and powerful for your .NET applications because of its machine learning features.